

# Simon Lin

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## EDUCATION

### Toronto Metropolitan University (Ryerson University)

Toronto, ON

*Bachelor of Science - Computer Science with Co-op*

*Sept. 2021 - present*

**Awards:** Dean's List (2021 - 2025)

**Relevant Courses:** Software Engineering, Data Structures, Artificial Intelligence, Web Development, OS

**GPA:** 4.12/4.33

## EXPERIENCE

### Software Engineer Intern

May 2025 – December 2025

*Magna International*

*Newmarket, ON*

- Developed and deployed full-stack web application with **Python + Flask + Vue.js** to accelerate **Finite Element Analysis (FEA)** processes to support **15+ global engineers**.
- Developed a **3D FEA model viewer** using **Three.js**, for visualizing Von Mises Stresses in complex mechatronic load conditions for vehicle hood latches and eliminated dependencies on expensive third-party design software.
- Built a CATIA automation tool, **reducing manual data entry** time by **70%** in **CAD** files and improving efficiency for cross-functional engineering teams.

### Software Engineer Intern

Apr. 2024 – Aug. 2024

*SOTI*

*Mississauga, ON*

- Developed new features for the MobiControl product line in **C#** and **.NET**, collaborating with product developers to translate technical specifications into production code.
- Implemented **REST APIs** to improve efficiency between client applications and backend services by **35%**.
- Designed and implemented **relational database schemas** using **Microsoft SQL Server**, normalizing data models to reduce redundancy and improve query performance across **5+ core product tables**.
- Collaborated within an **Agile CI/CD pipeline**, participating in weekly planning and code reviews to deliver features on schedule.

### Software QA Automation Intern

Sept. 2023 – April 2024

*SOTI*

*Mississauga, ON*

- Developed **automated** test plans, test cases and unit tests in **C#**, **Selenium** and **Microsoft Azure DevOps** adhering to software requirements and technical specifications for the MobiControl product line.
- Discovered and **resolved over 225 critical defects** and performance issues across product UI and dashboards.
- Lowered **SQL** database retrieval time by **27%** by optimizing legacy queries to clean, transform & aggregate data.

### Python Coding Instructor

July 2021 – Aug. 2021

*The STEAM Project*

*Richmond Hill, ON*

- Taught 20 summer camp students from grades 7-8 a creative curriculum centered on **Python** programming.
- Collaborated with co-instructors to create an educational and interactive coding curriculum on a weekly basis.
- Fostered engineering mindsets among students through daily coding challenges and projects.

## PROJECTS

**NBAction** | [Demo Video](#) | *Python, OpenCV, TensorFlow, PyTorch*

Sept. 2023 – Dec. 2024

*Basketball Action Classifier*

- Developed and trained a deep-learning model capable of detecting basketball actions to analyze live NBA games.
- Optimized a live video processing pipeline using dynamic frame sampling to reduce overall video latency by **68%**.
- Performed model fine-tuning to achieve an overall detection rate of **92%** across all object classes.
- Authored an [IEEE-style academic paper](#) detailing methodology and research results.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C#, JavaScript, TypeScript, Java, SQL, Bash

**Libraries & Frameworks:** Vue.js, React, Three.js, Flask, OpenCV, TensorFlow, .NET, Selenium, NumPy

**Developer Tools:** Git, Jira, Redis, Docker, Azure DevOps, GitHub